Would you take £50,000 or 50/50 chance of £1m?

Why people choose certainty over gamble

Option A: Guaranteed money

£50,000

You receive £50,000 immediately. There is no uncertainty.

Option B: A gamble

50%	50%
£1,000,000	**£0**
Winning Outcome	Losing outcome

You have an equal probability of winning or receiving nothing.

Reported survey responses

73%

Preferred the guaranteed £50,000



21%

Preferred the 50/50 chance of £1 million



6% are not accounted for by these two reported choices; do not assume their response category without checking the original survey.

Method -

1.Descriptive statistics - we use percentage 73%, 21% and 6% to summaries responses

2. Expected value - For fixed money - £50,000

For gamble - £1m *0.5 + 0 * 0.5 = £500,000

Key Result - 73 % people choose certainly over gamble

Takeaway- Risk preference reflect marginal utility and lose aversion, not only higher outcome.

Timestamp –

1) *00:43- 00:51* – Survey results: 73%, 21%, 6%.
2) *01:56* – Expected value calculation.
3) *02:47* – Behavioral economics and decision theory
4) *03:35–04:06* – marginal utility.
5) *04:26–06:21* – Prospect theory and loss aversion.
6) *06:33–08:16* – Wealth distribution and international comparison.

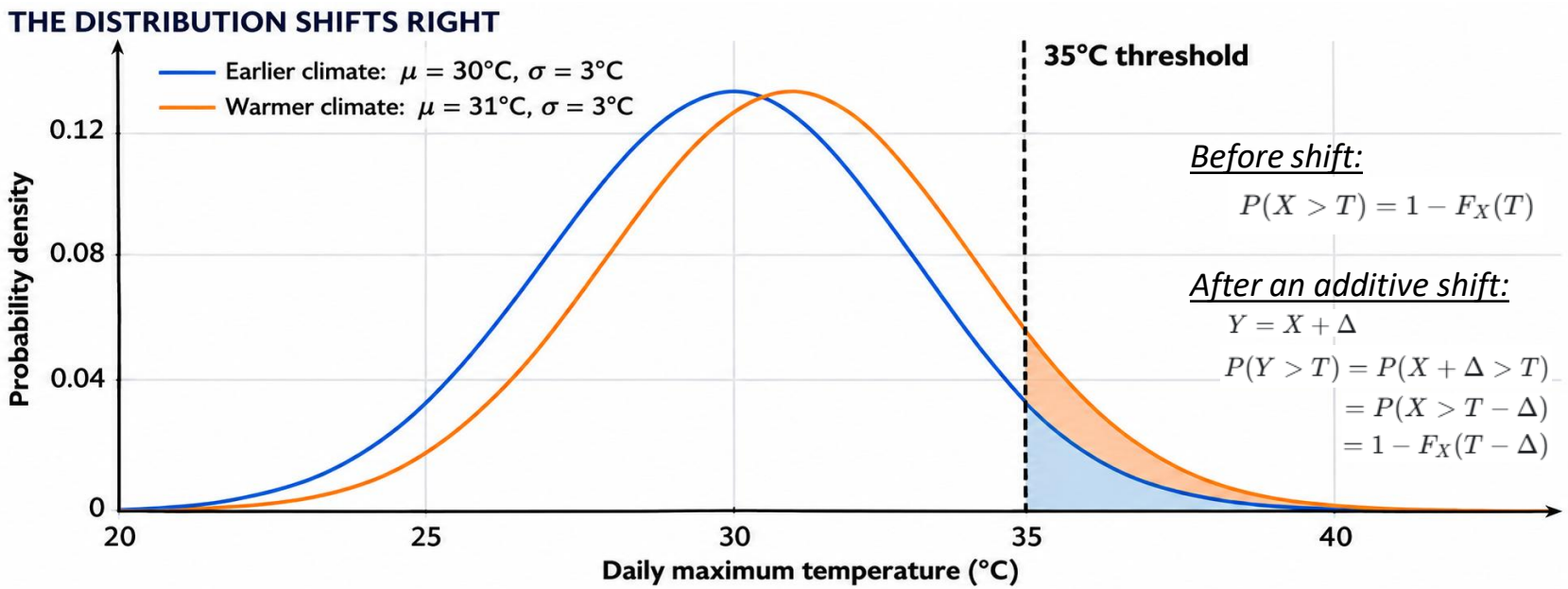
Source: BBC Radio 4, More or less, Published on 1 August 2026

Name- Vikas Yadav ,Student Id- MA24C045

Global Weirding: Why does a small temperature shift create many more heat waves?

"A Study of Tail behavior and Truncation"

Is studying average enough when there is small shift in every observation?



Blue = earlier climate; orange = warmer climate.

EARLIER CLIMATE
4.8% $P(T > 35°C)$ | MEAN (μ) **30°C** | z AT 35°C **1.67**

WARMER CLIMATE
9.1% $P(T > 35°C)$ | MEAN (μ) **31°C** | z AT 35°C **1.33**

Held constant: σ = 3°C | **Threshold:** 35°C

≈90%
RELATIVE INCREASE
in the probability of crossing 35°C

Why? With the spread unchanged, the threshold moves from 1.67 to 1.33 standard deviations above the mean. More of the distribution lies beyond 35°C.

Other Real-World Examples Where Tails Matter:

Boston: +3°F → 621% increase in days above 100°F.
Houston: +3°F → 466% increase in days above 100°F.

Boston Marathon speed +5% → 160% more runners finishing under 2:30.

Tropical storm wind speed +5% → Category 4 hurricanes +85%, Category 5 +350%.

Note: Hypothetical shifts applied to historical data.

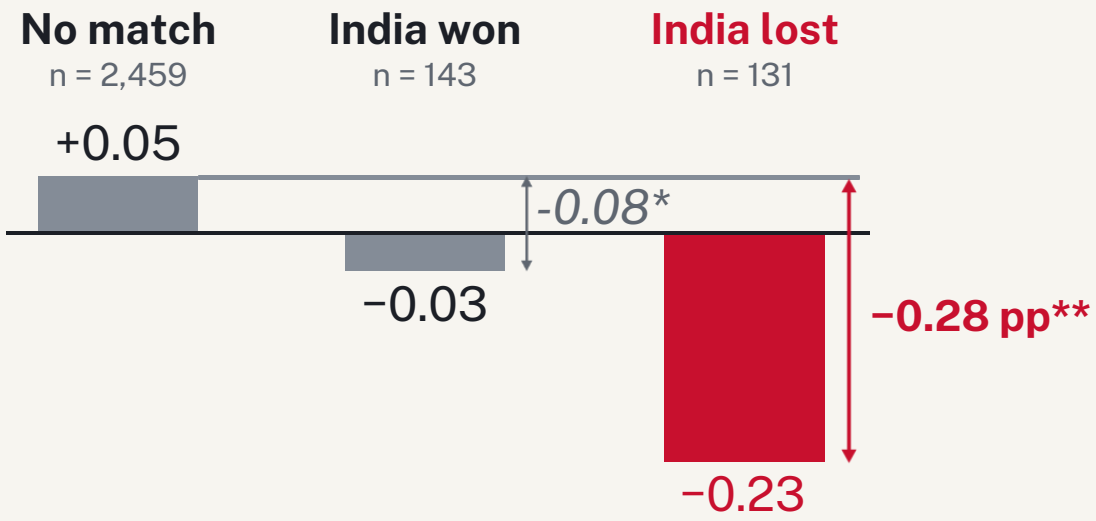
Conclusion:

- Small shifts in the distribution can sharply increase rare threshold exceedances.
- Averages alone can hide this risk.

Chronicles of Cricket: From Pitch to Portfolio

India stops for cricket — does its stock market share the mood?

AVERAGE NEXT-DAY NIFTY LOG RETURN (%), 1995–2005



$$R_t = 100 \ln\left(\frac{P_t}{P_{t-1}}\right)$$

METHODS

- $R_t = \beta_0 + \beta_W W_t + \beta_L L_t + \epsilon_t$
- *Stochastic-dominance test*

KEY RESULT

1.

	Coefficient	*t*-value
Victory	0.078*	0.63
Losses	−0.277***	−2.06
Losses (Sachin played)	−0.328***	−2.21

2. **Win $\succ_2$ Loss**
No match $\succ_2$ Loss

TAKEAWAY

Markets mourn cricket defeats but barely celebrate victories.

*- not significant; **-significantly different from 0 at the 10% level ***- significantly different from 0 at 5% level

Source: Mishra, V. & Smyth, R. (2008). "An Examination of the Impact of India's Performance in One-Day Cricket Internationals on the Indian Stock Market." Monash University, Dept. of Economics, Discussion Paper 09/08. Data: Table 1 (p. 29), Table 3 (p. 30), Table 4 (p. 31). Equations- (p.9), (p.11)

Do we stop finding new music at 31 or 33?

1 Evidence and methods

Which decade had the best music? [2]

Generation	1970s	1980s	1990s
Baby Boomers	**38%**	16%	3%
Gen X	17%	**38%**	18%
Millennials	10%	19%	**23%**

Weighted %. Selected groups/decades. Other responses omitted.
N = 17,578 (unweighted). Highlight: most selected decade per group.

Engagement: age vs listening time [3]

$r = -0.20$ $p < 0.001$

N = 9,080, ages 13–65, US and UK

Four listening items, summed. Pearson correlation.

Where do 31 and 33 come from?

Deezer [1,5]: peak discovery at 24, stagnation ≈31.
Self-reported behaviour in the UK survey.
Spotify [1,4]: median popularity ranks by age.
Parris interprets the early-thirties pattern as 33.

2 Interpretation, critique and applications

Three outcomes that are easy to confuse

Discovery
Music unfamiliar to the listener

Diversity
Range of artists
Most rotation: ages 25–34 [7]

Popularity
How mainstream the artists are

ILLUSTRATIVE EXAMPLE

Two listeners, age 40. Same obscure artist.

First encounter: discovery **Old favorite:** replay

Limits Age groups mix generations. Self-reports and shared accounts add error.

Better test Follow cohorts. Track artists absent from the previous 12 months of listening.

Application Learning platforms: unfamiliar subjects versus unpopular courses.

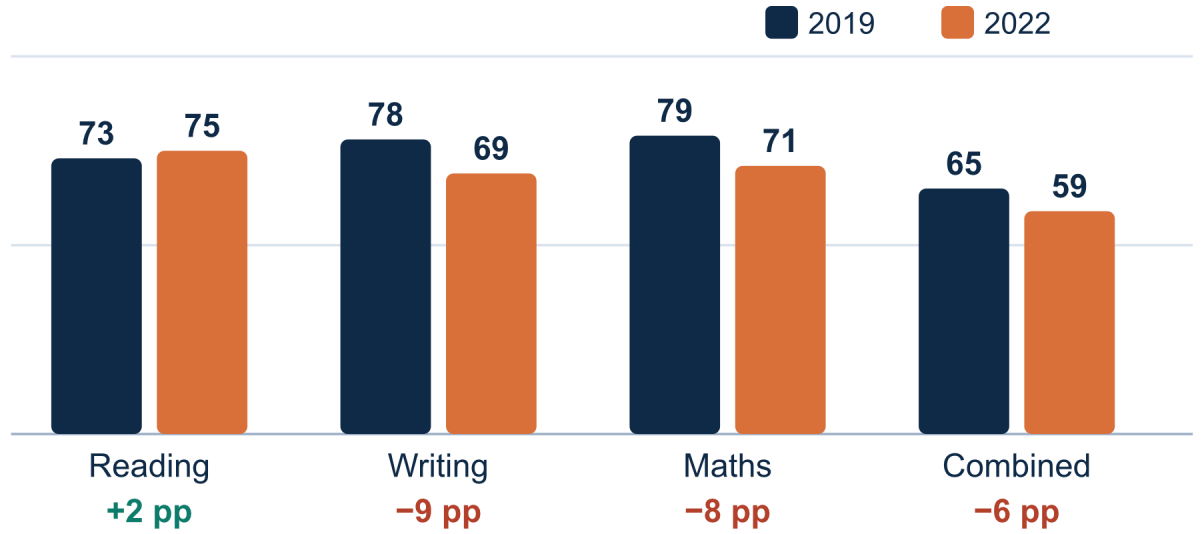
Age-related listening patterns do not establish a universal stopping age.

[1] Parris (2024), Stat Significant: “When Do We Stop Finding New Music?” and “Why Do We Stop Finding New Music?” sections.
[2] YouGov (2021), table p. 1. [3] Bonneville-Roussy et al. (2013), p. 705. [4] Kalia (2015). [5] Deezer via Dodgson (2018). [7] Lamere (2014). Full references in notes.

COVID-19 Disrupted Learning, Attendance and Wellbeing for Children in England

Official statistics show attainment fell, absence doubled, and device access shaped learning time.

Key Stage 2 pupils meeting the expected standard (%)



METHODS

Compared official before/after figures in percentage points; hypotheses tested at α = 0.05.

KEY RESULT

Combined attainment fell 65% → 59%; persistent absence nearly doubled to 21.2%.

TAKEAWAY

Associations, not proof of causation; targeted support for vulnerable pupils matters.

Persistent absence

10.9% → 21.2%

+10.3 pp (+94.5% relative)

Weekly study time, lockdown

8 h → 14 h

No device vs laptop: +6 h (+75%)

Probable mental disorder, H_0: prevalence equal (α = 0.05)

Ages 8–16: 12.5% → 17.1% (+4.6 pp) reject H_0

Ages 17–19: 25.7% → 23.3% (–2.4 pp) fail to reject H_0

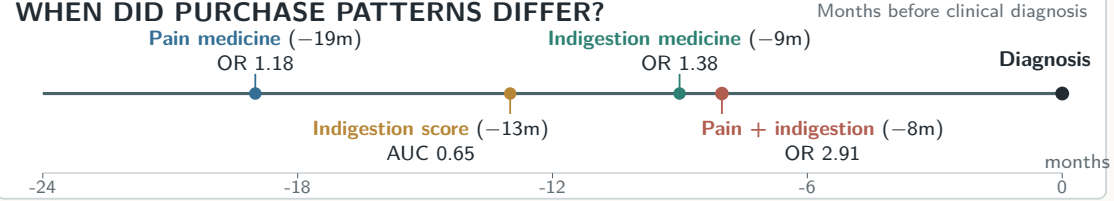
Podcast: [title], [outlet], [date]. Timestamps referenced: [mm:ss, mm:ss, mm:ss]. Data: Ofqual (2023); DfE (2024); House of Commons Library (2025); COSMO/Sutton Trust–UCL (2022); NHS England Digital (2023).

Can Your Loyalty Card Detect Ovarian Cancer?

Shopping patterns may contain early clues, but a statistical signal is *not* a medical diagnosis.

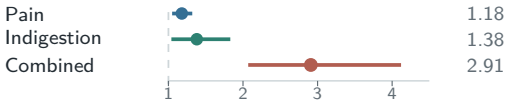
CLOCS STUDY 153 cases | 120 controls | two UK retailers | pain and indigestion medicines

WHEN DID PURCHASE PATTERNS DIFFER?



HOW STRONG IS THE ASSOCIATION?

OR 1 = no difference; bars show 95% confidence intervals



CAN PURCHASES PREDICT CANCER?

Indigestion score: 0.65 AUC (reported)



Some signal but not sufficient for diagnosis.

CAUTION Buying medicine is not the same as taking it and we need to consider selection bias and false alarms.

Media: BBC *More or Less*, 1 Feb 2023. **Study:** Brewer et al. (2023), JMIR 9:e41762, doi:10.2196/41762.
ORs use different models and AUC is not accuracy.

Group 38

Fair by one measure, unfair by another: auditing algorithmic risk scores

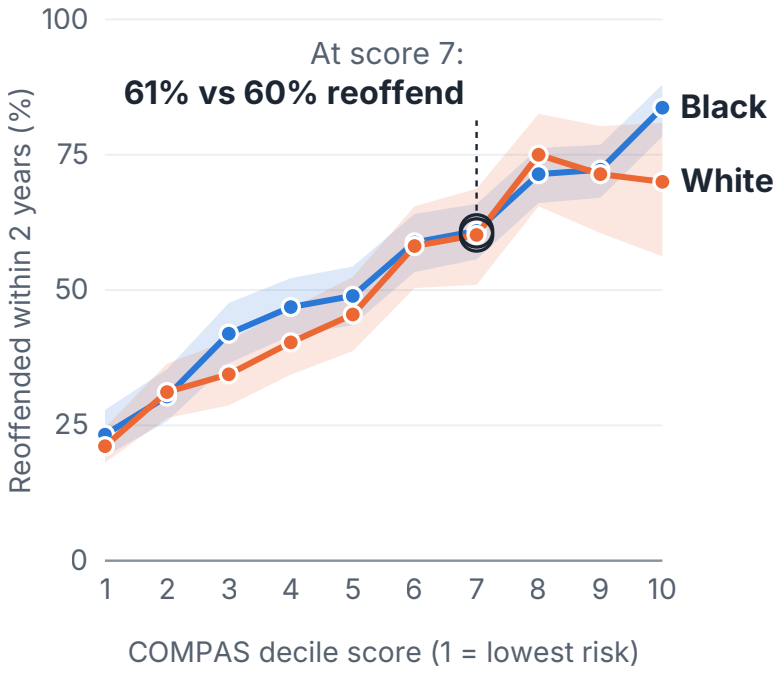
Roy John
ED26D007
MA 5104 Statistical Inference and Learning

Same score, same data, and two opposite verdicts on fairness.

(a) Calibration test

✓ Fair

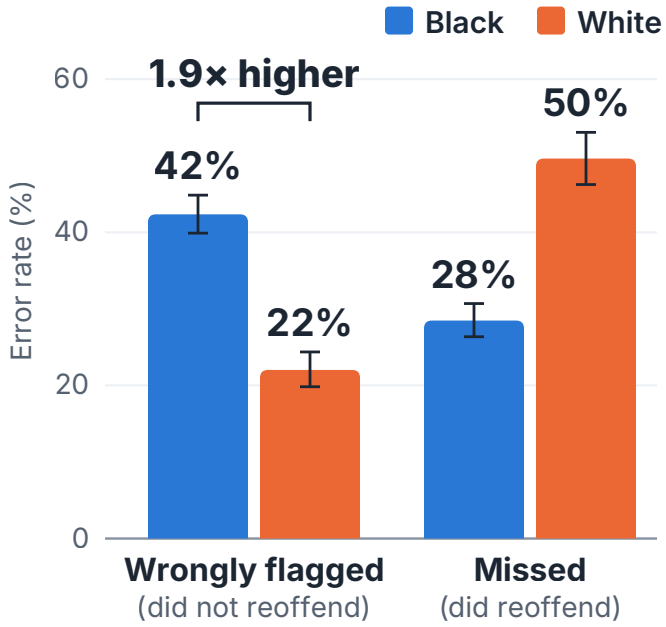
Does a score mean the same risk in both groups?



(b) Error-rate test

× Unfair

Are innocent people wrongly flagged equally?



Methods

Confusion matrices, calibration curves and z-tests on 5,278 Broward defendants.

Key result

Equal calibration, yet innocent Black defendants are flagged **twice as often**.

Takeaway

Choosing a fairness measure is a human decision, not a mathematical one.

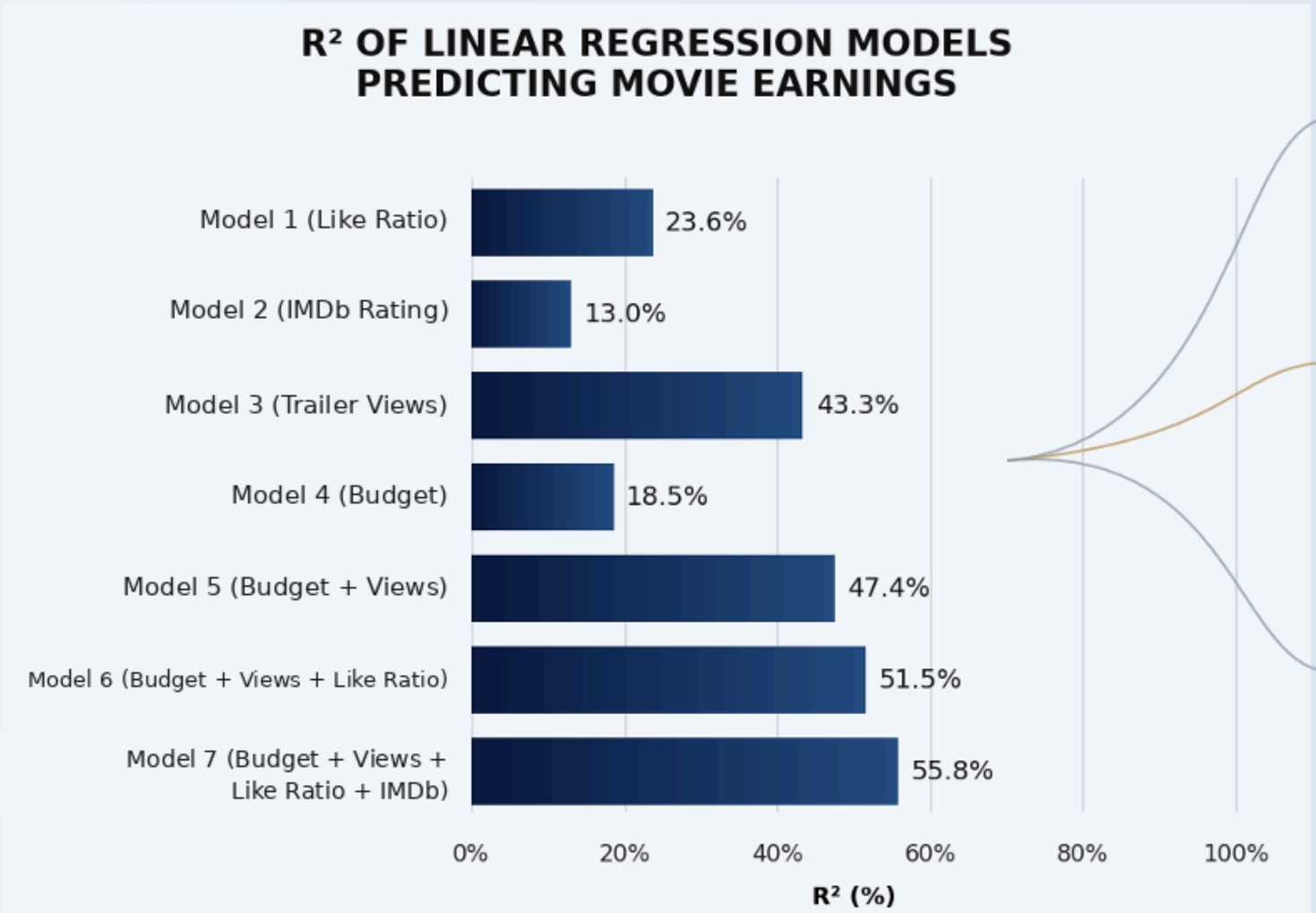
Media: Cathy O'Neil, "The era of blind faith in big data must end," TED2017, 2017, youtube.com/watch?v=_2u_eHHzRto. Key moments: **00:44, 01:48, 06:31, 07:38–08:12, 10:22–10:38**.

Data: ProPublica COMPAS, Broward County, FL, 2013–14 (n = 3,175 Black, 2,103 white defendants). High risk = decile ≥ 5; outcome = new charge within 2 years. Bands and whiskers: 95% Wilson CIs.

Augastya Srivastava (MA26M006) | Yashashree Pund (MA26M014)

DO MOVIE TRAILERS ACTUALLY SELL TICKETS? TRACKING CORRELATION AND CAUSATION!

A correlation of 0.66 links trailer views to earnings, but does that mean trailers sell tickets?



METHODS:
Regressed earnings on trailer views, likes, budget, and IMDb for 60 films.

KEY RESULT:
Trailer views explain 43% of earnings variation, most strongly for low- and mid-budget films.

TAKEAWAY:
Trailers track earnings, but the link varies by group; causation remains unproven.

MEDIA: Do Movie Trailers Actually Sell Tickets? - by Daniel Parris Sep 02, 2026.

Does praying to God affect recovery or survival in patients

Specifically: Effect of Intercessory prayer in cardiac bypass patients

Hemachandra MA25C019

Data Collection

Patients from 6 US Hospitals

1,802 patients participated

Control Groups

1. Unaware & being prayed for
2. Unaware & not being prayed for
3. Aware and being prayed for

Population Parameter

p = Proportion of population with complications up to 30 days post surgery

Claims being tested

1. Does intercessory prayer affect post surgery complications (p1/p2).
2. Does knowledge of being prayed for affect post surgery complication (p3/p1).

Results

1. p1 = 0.52, p2 = 0.51, p3 = 0.59
2. G1 vs G2: Relative Risk=1.02, 95% CI=[0.92,1.15]
3. G3 vs G1: Relative Risk=1.14, 95% CI=[1.02,1.28]

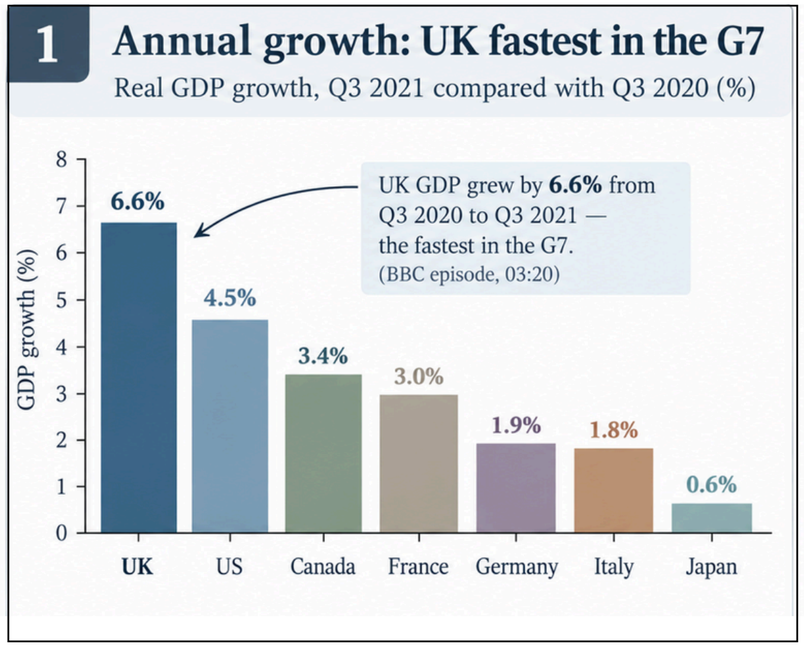
Conclusion

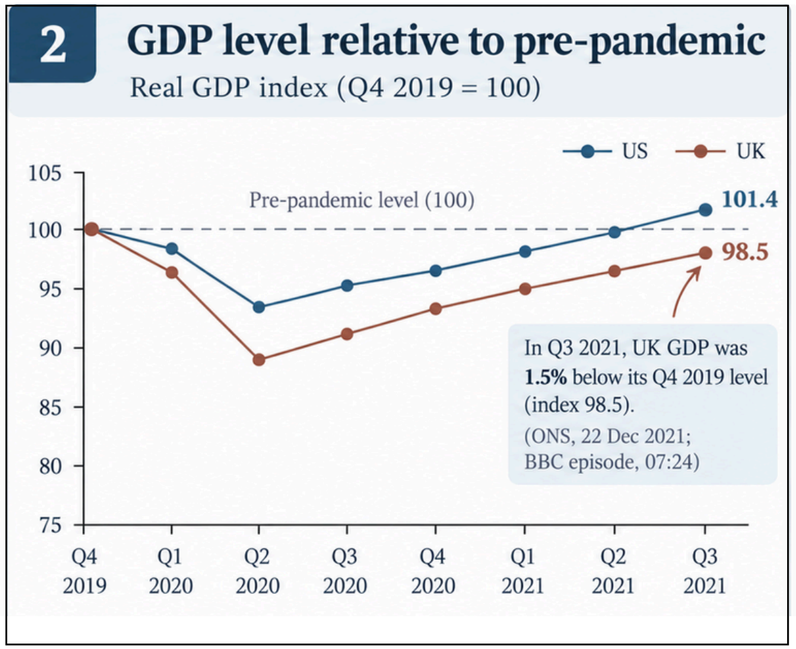
Neither intercessory prayer nor the knowledge of being prayed for positively affects the patients.

Source: Study of the Therapeutic Effects of Intercessory Prayer (STEP) in cardiac bypass patients: A multicenter randomized trial of uncertainty and certainty of receiving intercessory prayer

UK Growth in the G7: A Question of Measurment

A country can lead on **growth** and still **leg in recovery**.





GDP Growth Formula

Measures the percentage change in real GDP between two periods.

$$\text{GDP Growth (\%)} = \frac{GDP_t - GDP_{t-1}}{GDP_{t-1}} \times 100$$

GDP_t : Real GDP in the current period (t)

GDP_{t-1} : Real GDP in the previous period (t – 1)

- **Methods**: Comparative GDP quarterly analysis, pre-pandemic baseline tracking, and representative infection survey sampling.
- **Takeaway**: Statistical conclusions depend entirely on selected baselines, measurement periods, and definitions
- **Key result:** UK achieved 6.6% growth year-over-year, yet remained below pre-pandemic economic levels.

G7 Countries

- United States of America
- United Kingdom
- Canada
- France
- Germany
- Italy
- Japan

- Source: More or Less: Behind the Stats — “Does the UK Have the Fastest-Growing Economy in the G7?”, BBC Radio 4 / World Service, 2 February 2022.
- Key timestamps: 01:14–01:41 [Introduction]; 03:20 [GDP growth]; 06:38–07:24 [comparison period and pre-pandemic baseline]; 22:05–26:40 [Measurments for Growth Prediction].